

Basic Geometry

using pt. $\cong$ vector, so n being $\vec{n}$.

→ Dot Product

→ Plane Intersection

→ Cross Product

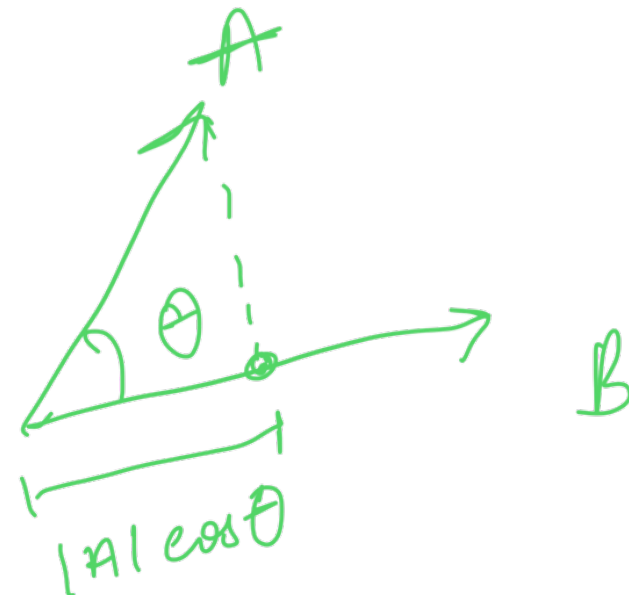
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→ Line Intersection

Dot Product

↳ Product
 ↳ Projection of 2nd vector onto the first one

$$a \cdot b = |a||b|\cos\theta$$



Properties

$$\hookrightarrow a \cdot b = b \cdot a$$

$$\hookrightarrow (\alpha a) \cdot b = \alpha(a \cdot b)$$

$$\hookrightarrow (a+b) \cdot c = a \cdot c + b \cdot c$$

$$\hookrightarrow \text{Norm of } a \rightarrow |a|^2 = a \cdot a$$

$$\hookrightarrow \text{Length is } \sqrt{a \cdot a} = |a|$$

$$\hookrightarrow \text{Proj of } a \text{ onto } b = \frac{a \cdot b}{|b|}$$

$$\hookrightarrow \text{Angle b/w vectors are } \cos\left(\frac{a \cdot b}{|a| \cdot |b|}\right)$$

↳ Angle

acute	- +ve
right	0
obtuse	- -ve

→ orthogonal dot product

independent of dimensions

$n-1$ dimensional subspace that divides n^{th} D space in 2 parts

Final property

Defining a hyperplane as $x \cdot n = C$, where $x \rightarrow$ set of points, n is a vector orthogonal to the hyperplane & C a fixed constant.

2D $\rightarrow$ Line

3D $\rightarrow$ Plane & so on...

$$x \cdot n = C \text{ or } (x - x_0) \cdot n = 0$$

where $x_0 \cdot n = C$

i.e x_0 lies on the hyperplane.